The empirical recurrence for OEIS A234927, proved by transfer matrix

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Abstract

OEIS A234927 counts arrays of width 3 over $\{0, 1, 2\}$ in which every 3×3 subblock satisfies a condition on its nine entries. The entry carries an empirical recurrence of order 6 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical: a 3×3 block lies in three consecutive rows, so the state is the last two rows and the count is a walk count on $S = 67$ states.

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1 The conjecture

OEIS A234927 is "Number of (n+2) X (1+2) 0..2 arrays with each 3 X 3 subblock having the sum of its 72 absolute element differences equal to 34 and no adjacent elements equal.". It has offset 1 and begins

$$56, 88, 138, 176, 312, 432, 584, 960, \dots$$

The entry states, as a conjecture contributed by its author and never marked settled:

> Empirical: a(n) = 2*a(n-3) + 4*a(n-6) for n>9.

As of the "Last modified" line on the live entry (Oct 16 2018 11:25:06, revision 8) this is still recorded as empirical, and nothing on the entry records it as proved.

2 What is being counted

The arrays have 3 columns and a growing number of rows, with entries in $\{0, 1, 2\}$. The condition is imposed on every 3×3 block of the array — every choice of three consecutive rows and three consecutive columns — and asks that the sum of the absolute differences over the 36 pairs of distinct entries equals 34. The entry adds that no two adjacent entries of the array are equal.

The entry counts those differences as 72, which is the number of ORDERED pairs; the number it compares against is the sum over the 36 unordered pairs, and that is the reading its published terms confirm — the value named is not an achievable ordered-pair sum at all.

3 Three rows

A 3×3 block occupies three consecutive rows, so everything the condition asks is decided by three consecutive rows and nothing wider. Take the array one row at a time and let the state be the last two rows written, with a distinguished start standing for the array before its first row. Appending a row completes the blocks whose bottom row is the new one, and the step is allowed exactly when each of those blocks passes.

Every block of the finished array is tested exactly once, at the step that writes its bottom row, and no step imposes anything the definition does not ask. The first two rows complete no block at all, which is why they are laid down without a test. Every state may end an array, so $\tau = \mathbf{1}$. Thus

$$a(n) \;=\; \iota^{\top} M^{j} \tau, \qquad S = 67,$$

for the transfer matrix M on those S states. The walk index j and the entry's own n differ by a constant, read off by matching the published terms rather than assumed. In particular a is C-finite.

4 The proof

Lemma 1. *Let $q(t) = t^6 - \sum_i c_i t^{6-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^{\top} M^{j} q(M) \tau$ for the corresponding walk index j, so the recurrence holds from some point on if and only if $\iota^{\top} M^j q(M)\tau = 0$ for all large j, and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+6} - \sum_i c_i M^{j+6-i}$. For the bound, M^S is an integer combination of $I, M, \ldots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^{\top} M^j q(M)\tau$ obey the monic recurrence given by the characteristic polynomial of M; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. □

Theorem 2.

$$a(n) \;=\; 2\,a(n-3) + 4\,a(n-6)$$

for every $n > 9$.

Proof. The vector $w = q(M)\tau$ was formed by 6 matrix–vector products in exact integer arithmetic and $\iota^{\top} M^j w$ evaluated for $j = 0, 1, \ldots,$ again exactly; those integers vanish from the index corresponding to $n = 9 + 1$ onwards and the run continued until the working vector was identically zero, which settles every larger j at once. □

5 Verification

Four checks, each independent of the algebra above.

First, the model reproduces all 33 terms the entry publishes, exactly, in integer arithmetic.

Second, the count was recomputed for the smallest arrays straight from the definition: every array was written out cell by cell and each 3×3 block tested directly on its nine entries. No rows, no window, no state and no transfer matrix are involved in that computation, and the published terms came back.

Third, the conjectured recurrence was evaluated directly on the published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Fourth, the annihilation test of Lemma 1 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A234927.

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[3] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.